MIL-D-24483A(SHIPS)
19 August 1974
SUPERSEDING
MIL-D-24483(SHIPS)
8 May 1973
(See 6.8)

MILITARY SPECIFICATION

DECK COVERING, SPRAY-ON, NONSLIP

§ This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

§ 1.1 Scope. This specification applies to a spray-on, nonslip, lightweight deck covering for metal (or wood) decks.

§ 1.2 Classification. The total deck covering system consists of a primer and a nonslip topcoat (base resin, nonslip aggregate, and hardener). Topcoats shall be furnished in the following types, as specified (see 6.2):

Type I - General purpose.
Type II - Aircraft carrier landing area.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-5572 - Gasoline, Aviation: Grades 80/87, 100/130, 115/145.
MIL-H-5606 - Hydraulic, Fluid, Petroleum Base; Aircraft, Missile and Ordnance.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-6091 - Alcohol, Ethyl, Specially Denatured, Aircraft.
MIL-D-16791 - Detergents, General Purpose (Liquid), Nonionic.
MIL-G-18709 - Grease, Ball and Roller Bearing.
MIL-P-24441/1 - Paint, Epoxy - Polyamide, Green Primer, Formula 150.

FEDERAL

QQ-W-470 - Wire, Steel, Carbon, Spring, Music.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 thru 12 Gallon).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.
§ MIL-STD-1623 - Fire Performance Requirements and Approved Specifications For Interior Finish Materials and Furnishings (Naval Shipboard Use).

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.
FED-STD-313 - Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials.
FED-STD-595 - Colors.
FED-STD-601 - Rubber: Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D1141-52 - Substitute Ocean Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 "P" Street, N.W., Washington, D.C. 20036.)

LOS ANGELES AIR POLLUTION CONTROL REGULATIONS
Rule 66 - Organic Solvents

(Application for copies should be addressed to the Los Angeles County Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, California 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The deck covering materials furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Description. The material furnished under this specification, when mixed and applied by spray in accordance with the manufacturer's instructions, shall produce a topcoat with uniformly distributed nonslip aggregate which shall protrude prominently above the resin matrix.

3.3 Composition. The deck covering "system" shall consist of a primer and a topcoat, the latter to include a nonslip aggregate. The primer shall conform to MIL-P-24441/1, Formula 150. The topcoat shall be a two part synthetic resin compound consisting of the base resin plus aggregate, and hardener or curing agent.

3.4 Condition-in-container. The components of the topcoat shall be readily broken up with a paddle to a uniform consistency, shall not liver, gel, or show any other objectionable properties which will affect performance, and shall produce a topcoat which meets all the requirements of this specification for at least one year after date of delivery (see 4.6.2).

3.5 Nonvolatile content. When tested as specified in 4.6.3, the minimum nonvolatile content of the topcoat deck covering shall be 88 percent.

3.6 Pot life. When tested as specified in 4.6.4, the minimum pot life shall be 4 hours.

3.7 Thickness. When tested as specified in 4.6.5, the average overall minimum thickness, including aggregate, when cured, shall be 30 mils.

3.8 Weight. When tested as specified in 4.6.5, the weight shall be not more than 4.0 ounces per square foot.

3.9 Coverage. When applied as specified in 4.6.6, the minimum coverage shall be 60 square feet per gallon.

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3.10 Fire resistance. The deck covering shall conform to the fire test requirement set forth in MIL-STD-1623 (see 4.6.14).

3.11 Resistance to impact. When tested as specified in 4.6.7, the deck covering shall have a minimum rating for all tests of 95 percent except for impact after seawater immersion where the minimum rating shall be 30 percent.

3.12 Abrasion of arresting gear cable (type II only). When tested as specified in 4.6.8, the type II covering shall produce a maximum average wear of the test wire of 0.4 mils.

3.13 Nonslip properties. When tested as specified in 4.6.9, the coefficients of static friction for the deck covering shall not be less than the values shown in table I.

Table I - Coefficient of static friction (minimum values).

Test condition	Rubber surface		
	Dry	Wet	Oily
Initial	1.15	1.00	0.95
After wear test	1.10	1.00	0.95
After cable wear test (Type II only)	1.00	0.90	0.50

3.14 Resistance to wear. When tested as specified in 4.6.10, the deck covering shall produce a maximum reduction in thickness of 25 mils. In worn areas, the evenly distributed aggregate should continue to protrude above the resin matrix.

3.15 Resistance to accelerated corrosion. When tested as specified in 4.6.11, the deck covering system shall not become detached. On the 6-inch panel, beyond the 1/4-inch radius from the center of impact (see 3.11), there shall be no separation at the primer-topcoat interface on the non-impact areas.

3.16 Resistance to moisture and temperature changes. When tested as specified in 4.6.12, the deck covering system shall show no signs of cracking, corrosion of the steel substrate, or separation of the primer-topcoat interface.

3.17 Resistance to immersion. When tested as specified in 4.6.13, the deck covering system shall show no signs of softening, loss of adhesion at the steel-primer and primer-topcoat interfaces, discoloration, or any other form of deterioration.

3.18 Flash point. When tested as specified in 4.6.15, the mixed ingredients shall have a minimum flash point of 80° Fahrenheit(F).

3.19 Drying time. When the components of the topcoat are mixed and applied as specified in 4.6.1, the topcoat shall have a maximum drying time of 24 hours.

3.20 Color. Topcoats shall be available in the following colors: yellow to match No. 33538, red to match No. 31136, olive drab to match No. 24087, haze gray to match No. 36270, or dark gray to match No. 36076 of FED-STD-595; or dull black or white (with minimum reflection of 55 percent (see 4.6.16 and 6.2)).

3.21 Identification characteristics. Values for identification characteristics shall be provided by individual suppliers for characteristics as indicated by "X" in table II. The purpose of these values is to serve as a basis for determining that the material being offered is essentially the same as that which was approved under qualification testing.

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Table II - Identification characteristics values.

Characteristic	Base material, component A	Curing agent, component B	Combined mixture	Aggregate
Chemical nature	X	X	---	X
Percent of each principal constituent (5 percent or more of total)	X	X	---	
Percent pigment plus aggregate (if applicable)	X	X	---	
Percent volatile	X	X		
Percent non-volatile vehicle	X	X		
Weight per gallon	X	X	X	
Consistency	X	X	X	
Flash point	X	X	X	
Pot life				
at 90°F	---	---	X	
at 50°F	---	---	X	
Coverage	---	---	X	
Drying	---	---	X	
Mixing instructions	---	---	X	
Application instructions	---	---	X	
Aggregate sizes (sieve test)	---	---	---	X
Infrared spectrum (2 to 15 micrometres)	X	X		
Epoxy content (method 7403 of FED-STD-141)	X			
Compliance with Air Pollution requirements 2/		Certify 2/ Yes or No		

1/ If yes, provide data on solvent composition.

2/ Formulation must comply with the requirements of Rule 66 of the Los Angeles Air Pollution Control District (January 28, 1966).

3.21.1 Values shall be established for each topcoat and shall be submitted with the application for qualification testing. The contractor shall also furnish a copy of the DD 1813 (see 5.3.1.3) to the Naval Ship Engineering Center with the application for qualification testing.

3.22 Serviceability. The deck covering systems shall show no loss of adhesion or other deficiency which would limit their serviceability when examined after a minimum of six months for type I or II material, or after a minimum of 1500 landings for type II material (see 4.6.17).

3.23 Service use. The type I deck covering system shall continue to provide satisfactory performance without excessive touch-up for a minimum period of one year (see 4.6.17).

3.24 Directions for mixing and spraying. Written directions for mixing and spraying as recommended by the manufacturer, including working life, application temperature limitations, and safety precautions, shall be furnished with each contract, as specified in 6.2. Directions which differ from those furnished for qualification testing under 3.21 shall be submitted to the Naval Ship Engineering Center for review.

3.25 A tabulation of requirements and related sections of this specification is contained in table III.

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Table III - Requirements.

Item	Test requirement	Test method	Applicable paragraph
Drying time, hours, maximum	24	4.6.1	3.19
Condition in container	shall pass test	4.6.2	3.4
Non-volatile content, percent, minimum	88	4.6.3	3.5
Pot life, hours, minimum	4	4.6.4	3.6
Thickness, mils, minimum	30	4.6.5	3.7
Weight, ounces/sq. ft., maximum	4.0	4.6.5	3.8
Coverage, sq. ft./gallon, minimum	60	4.6.6	3.9
Fire resistance:			
Average char, inches, maximum	see MIL-STD-1623	4.6.14	3.10
After-flaming	None		
Impact resistance:			
Initial and after aging percent, minimum	95	4.6.7	3.11
After seawater immersion, percent, minimum	30	4.6.7	3.11
Abrasion resistance of cable, mils, maximum (type II only)	0.4	4.6.8	3.12
Non slip properties	See table I	4.6.9	3.13
Wear resistance, mils, maximum	25	4.6.10	3.14
Corrosion resistance	shall pass test	4.6.11	3.15
Moisture and temperature resistance	shall pass test	4.6.12	3.16
Immersion resistance	shall pass test	4.6.13	3.17
Flash point, degrees F, minimum	80	4.6.15	3.18
Color	to match	4.6.16	3.20
ID characteristics	see table II	----	3.21
Serviceability	shall pass test	4.6.17	3.22
Service use	shall pass test	4.6.17	3.23

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing shall be as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of all tests specified in this specification. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3). Requalification will be required every three years, and shall consist of conducting all tests, except service test and one year condition in container test. Service test shall be conducted on a carrier flight deck in accordance with 4.6.17. Stability test data required by 4.6.2 shall be furnished within one year from date of application for qualification.

4.3.1 Ingredient samples. For qualification, the supplier shall submit to the designated laboratory, one pint container of the aggregate ingredient, one gallon container of the base material, component A (with aggregate as applicable), and one gallon of curing agent, component B, each packaged in a separate container.

4.3.2 Qualification approval for type II will also constitute approval for type I; however, approval for type I does not constitute approval for type II.

4.4 Quality conformance inspection.

4.4.1 Lot. For purposes of examination and testing, a lot shall consist of a production batch of topcoat that is offered for delivery under a contract or order.

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4.4.1.1 Sampling for quality conformance inspection.

4.4.1.1.1 Sampling for examination of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level = 2.5 percent defective for the examination specified in 4.4.2.

4.4.1.1.2 Sampling for quality conformance tests. From each lot of material for delivery, a two-quart sample of base material with sufficient curing agent, shall be selected at random for the quality conformance tests specified in 4.5. Where lots are small, only one 2-quart sample of base material shall be selected for 500 gallons of material.

4.4.1.1.3 Sampling for verification tests. From the first lot of a single type offered for delivery under a contract or order, and for every 5,000 gallons thereafter, one 2-quart sample of base material with sufficient curing agent, shall be selected at random and forwarded to a designated Government laboratory for verification tests.

4.4.2 Examination. Each filled container in the sample selected in accordance with 4.4.1.1.1 shall be examined to verify compliance with this specification. Any container in the sample having one or more defects under required fill shall not be offered for delivery. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105 (see 4.4.1.1), this shall be cause for rejection of the lot represented by the sample.

4.4.3 Quality conformance tests. Quality conformance tests for acceptance of individual lots shall consist of all tests specified in 4.5.1. A copy of the quantitative test data (see 6.2.1) for each lot, noting contract number and date of manufacture, shall be forwarded to the Naval Ship Research and Development Center (Code 2841), Annapolis, Maryland 21402.

4.5 Acceptance of individual lots. Acceptance of individual lots shall be based on compliance of material with the desired results obtained by 4.5.1.

4.5.1 Lot acceptance tests. The following tests shall be performed and results shall meet the applicable requirements in section 3.

Drying time	4.6.1
Initial condition in container	4.6.2
Non-volatile content	4.6.3
Pot life	4.6.4
Thickness	4.6.5
Weight	4.6.5
Resistance to impact	4.6.7.1(a), (b), and (c)
Nonslip properties (initial)	4.6.9
Flash point	4.6.15
Color	4.6.16

4.6 Test procedures.

4.6.1 Preparation of specimens. Specimens of the size specified in the following tests shall be prepared, using 1/8 inch thick mild steel plates except for the 0.0625 inch thick mild steel plates used for thickness and weight as specified in 4.6.5, and for nonslip and wear as specified in 4.6.9 and 4.6.10 respectively. Panels shall be blasted to a uniform white appearance with a minimum surface profile of 1.0 - 2.0 mils peak to valley. Panels shall be cleaned after blasting, using clean, dry compressed air, vacuum, or solvent mixture of 1:1 xylene and isopropanol. Material shall be mixed and sprayed in one coat in accordance with the procedure recommended by the manufacturer. The coated steel specimens shall cure for a minimum of 96 hours before conducting tests. Unless otherwise specified herein, all curing and testing shall be conducted under atmospheric conditions at a temperature of 72° ± 3°F, and a relative humidity of 50 plus or minus 2 percent.

4.6.1.1 Drying time. The dry-hard time shall be determined by method 4061.1 (dry-hard) of FED-STD-141.

4.6.2 Stability.

4.6.2.1 Storage. The manufacturer shall certify and shall support his certification with test data that all components of deck covering systems after being sealed in their original containers and held in storage for a period of 1 year, between 50° to 90°F, shall conform to 3.4. Test data shall consist of results of tests specified in 4.5.1.

4.6.3 Nonvolatile content. The nonvolatile content of the compound shall be determined in accordance with method 4041 of FED-STD-141.

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4.6.4 Pot life. The packaged ingredients comprising the components of the topcoat shall be thoroughly mixed by hand and allowed to stand in an open container at a temperature of 75 plus or minus 3 degrees Fahrenheit and a relative humidity of 50 plus or minus 2 percent. The components shall be placed in a quart container in sufficient quantity to fill the container to approximately 1/2 inch from the top. Mixing procedures shall be in strict accordance with the manufacturer's instructions. After 4 hours, the material shall be sprayable in accordance with manufacturer's instructions.

4.6.5 Thickness and weight. The thickness and weight of three 6-inch by 6-inch square, 0.0625 inch thick, clean, mild steel plates shall be measured. The weight shall be determined to the nearest 0.1 gram. The thickness shall be determined at 16 equally distributed points on each of the plates by means of a dial type thickness gage and template. The material shall be sprayed one coat in accordance with the manufacturer's instructions and allowed to cure for a minimum of 96 hours. After curing the three test specimens, each shall be weighed to the nearest 0.1 gram. The length and width shall be measured to the nearest 0.01 inch and the thickness to the nearest 0.001 inch at each of the points previously measured. The difference in thickness of the uncoated steel plate and the coated steel plate at each of the points shall be determined and the results averaged for each of the specimens. The thickness of each specimen shall then be averaged and this value reported as the coating thickness. The weight of the coating shall be determined from the difference between the weight of the coated steel plate and the uncoated steel plate. The weight of the three specimens shall be averaged and multiplied by four to convert to weight per square foot.

4.6.6 Coverage. Coverage shall be determined by applying the topcoat to a 3 foot by 3 foot by 1/8-inch thick mild bare steel plate in accordance with the application procedure recommended by the manufacturer to produce mixing and a minimum thickness of 30 mils (see 4.6.5). Result shall be the average of at least two determinations.

4.6.7 Resistance to impact.

4.6.7.1 Procedure. A set of three specimens, 6 by 6 inches, shall be prepared as specified in 4.6.1 to represent each of the treatments below. Topcoat shall be applied over one spray coat of epoxy primer conforming to MIL-P-24441/1, formula 150. Dry film thickness of epoxy primer shall be approximately three mils.

(a) No treatment, or "as is"
(b) 4 hours at 32°F
(c) 4 hours at 158°F
(d) 100 hours weather aging and intermittent water spray in an Eveready X-1a unit or equal in accordance with method 7311 of FED-STD-601.
(e) 15 day immersion in synthetic seawater.

4.6.7.2 Immediately upon removal from treatment, the specimens shall be subjected to 25 impacts by a 2-pound solid steel ball dropped from a height of 5 feet. The steel ball shall be held in a fixed position by an electromagnet provided with a centering tip. The specimen shall be held on a 1-1/2 inch thick steel base which shall be fitted with guides to reposition the specimen after each impact, so that successive points-of-impact are equally spaced 3/4 inch apart for 25 impacts in a 5 by 5 pattern (5 rows of 5 impacts per row) enclosed within an area of approximately 9 square inches.

4.6.7.3 Rating. Upon completion of the impact test, each specimen shall be examined for failure of the covering between impacts. The evaluation will be based on a percent estimate of the portion of the covering between impact sites that remain intact. A rating of 100 percent would show 25 discrete impacts with the covering around each impact firm and adhering to the substrate. A rating of 0 percent would show all the covering removed within the 9 inch square area.

4.6.8 Arresting gear cable wear (type II only). For type II topcoat, three specimens 8 inches long by 4 inches wide shall be prepared as specified in 4.6.1, over one spray coat of epoxy primer, formula 150 (approximately 3 mils dry film thickness). Each specimen shall be subjected to 200 cycles of wear by an 1/8 inch diameter music wire conforming to QQ-W-470. Each specimen shall be evaluated with a new wire. The diameter of each wire shall be

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measured. The wire shall then be held in a jig which is clamped in the test apparatus as shown on figure 7. After completion of 200 cycles in the wear test machine, the wire shall be removed from the jig and the amount of wear on the wire shall be determined by measuring the thickness of the wire. Measurements shall be made to the nearest 0.0001 inch by means of a dial type thickness gage having a foot fitted with a knife edge which shall contact the worn surface of the wire, and a 1 inch diameter flat anvil on which the wire shall be supported. The average of ten measurements for each three wires shall be taken as the average wear caused by the topcoat.

4.6.8.1 Arresting gear cable wear test apparatus. The test apparatus as shown on figure 1 shall have essential characteristics as follows:

(a) A reciprocating carriage with a total throw of 6 inches per cycle, a frequency of approximately 30 cycles per minute and shall be equipped with clamps to hold the 8 inch by 4 inch steel plate coated with the test material.

(b) A pivoted clamp assembly arranged to support a jig for holding an eight inch length of 1/8 inch diameter music wire with its axis horizontal and at right angle to the direction of motion of the reciprocating carriage. The jig shall consist of an eight inch long, one inch diameter steel bar into which a 1/6 inch "V" notch has been cut along its length for holding the wire in contact with the nonslip deck coating without flexing or twisting. The pivot shall be 12-1/2 inches from the wire held in the "V" notch of the jig, and the entire pivoted assembly (including the test wire and jig) shall be weighted such that the wire shall contact the coated specimen under a load of 20 and 1/4 pounds. The test wire in the jig shall be clamped at each end so that no part of the coated clamp contacts the coated specimen.

4.6.9 Nonslip properties (coefficient of static friction). A total of three specimens, 2 inches by 3 inches, shall be prepared as specified in 4.6.1. These specimens shall be used in the evaluation of initial static friction over dry and synthetic seawater wet surfaces. After 225 revolutions on the wear test machine, the specimens shall be evaluated for static friction. An additional three specimens shall be prepared as specified in 4.6.1, for determination of initial static friction on an oily surface. Wet the specimen with synthetic seawater prepared from sea salt in accordance with ASTM D1141-52 to produce the wet surface. The specimens shall then be rinsed in tap water to remove the synthetic seawater, blotted with paper towels, dried in an oven at 250°F for 1 hour, and then permitted to cool to ambient temperature prior to testing at the oily surface condition obtained by wetting with SAE 10W oil at a temperature between 60° and 80°F.

4.6.9.1 Procedure. The coefficient of static friction shall be determined by means of the Olsen Slipmeter, as shown on figure 2. The slipmeter is a portable instrument with a 6 pound weighted meter that is supported by a set of three removable, 3/4-inch diameter feet. Two sets of feet shall be fitted with vulcanized neoprene rubber having a Shore "A" hardness range of 60-80 durometers. Data shall be obtained by pulling the slipmeter over the specimens at a rate of 4.0 inches per minute by means of a cord that is wound over a constant speed motor-driven winch positioned in front of the specimens. A set of three 2-inch by 3-inch specimens shall be held along their 3-inch axis and arranged as shown on figure 3, such that each foot of the slipmeter contacts a separate specimen. For each of the three readings, the three specimens support a different foot of the slipmeter. The specimens are then rotated 180 degrees and the procedure repeated for a total of six readings. Each of the 4-inch by 8-inch specimens shall be held lengthwise so that all three feet of the slipmeter are in contact with the specimen during tests. One reading shall be recorded on each of the three 4-inch by 8-inch specimens after which the specimens shall be rotated 180 degrees and readings for each specimen shall be again recorded for a total of six readings. After wear by the wear test machine (2-inch by 3-inch specimens) (see 4.6.10), and by the arresting gear cable wear test apparatus (4-inch by 8-inch specimen) (see 4.6.8), the static friction shall be determined with the slipmeter for each of three surface conditions, dry, wet with synthetic seawater and oily. One set of rubber feet shall be used for the dry and wet surface conditions and another set of rubber feet shall be used for the oily surface condition.

4.6.9.2 Computation. The average of the recorded readings from the dial gage on the slipmeter shall be converted into friction load in pounds from the conversion curve on figure 4. The coefficient of friction shall be computed by dividing the average friction load by 6 pounds, which is the total weight applied to the three feet of the slipmeter.

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4.6.10 Resistance to wear. Three specimens 2-inches by 3-inches prepared as specified in 4.6.1, shall be mounted so that an area of approximately 6 square inches of the coating systems is presented to the abrasive on the disk of the machine described herein. The abrasive shall be number 80 aluminum oxide. The thickness shall be measured with a dial type thickness gauge at 12 equally distributed points positioned by use of a template. The specimen shall then be subjected to 225 revolutions of the wear test machine, and the thickness obtained at the template positioned points previously measured. The difference in thickness shall be averaged to determine the thickness of material worn away by the abrasive.

4.6.10.1 Wear-test machine. The wear-test machine as shown on figures 5 and 6 shall have the following essential mechanical characteristics:

(a) Mild steel, abrasion disk 14-1/8 inches in diameter and 1/2 inch thick, revolved clockwise at a constant speed of 23.5 revolutions per minute (r/min).
(b) Specimen holder, attached to a shaft. The distance between center of the disk and the specimen holder shaft shall be 5.1 inches. The holder shall revolve clockwise at a constant speed of 32.5 r/min. In each one-half revolution of the shaft, the specimen shall be lifted a distance of 1/16 inch and then dropped back on disk (a) under a 10 pound weight. Change in the lift due to wear of the specimen shall be compensated by adjustment of a cam at regular intervals, so that the lift is maintained at a practically constant distance.
(c) Weight, 10 pounds the maximum found suitable for use with this test machine.
(d) Distribution of abrasive grit. The grit flows by gravity from a hopper through a 13/16 inch inside diameter tube and into the center of a distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grit through eight 3/64-inch diameter holes centered on a 1-5/8 inch diameter pitch circle. A uniform distribution of grit shall fall to disk (a) in the path of the test specimen. The distributor shall be revolved at a constant speed of 9 r/min.

Details of the distributor for the abrasive grit, the cam and specimen holder are shown on figures 7 and 8 respectively.

4.6.10.2 Procedure. The steel disk (a) shall revolve while the weighted specimen of the material in holder (b) shall be lifted and dropped by a cam (c) while it is being revolved. When the specimen is dropped through motion of the cam, it shall fall a distance of 1/16 inch and meet disk (a) with impact simulating the fall of a foot on the deck surface. Foot action shall be further simulated in that the revolving specimen shall contact the disk with a sliding and twisting motion until lifted clear by the cam. Continuous rotation of the specimen shall introduce the element of wear in all directions on the material. Free abrasive grit of number 80 aluminum oxide shall be fed at a steady rate from the hopper and distributed uniformly on the steel disk in the path of the specimen. The revolving motions of the steel disk and the test specimen shall keep the abrasive grit circulating to the outer side of the path of the specimen. The grit shall fall from the edge of the steel disk into hopper from which it is collected and then screened to eliminate undesired particles before replacement in the upper feeding hopper.

4.6.11 Resistance to accelerated corrosion.

4.6.11.1 Apparatus. The apparatus consists essentially of a glass jar with a specimen rack, containing salt solution through which air is bubbled to accelerate corrosion. The glass jar shall be a nominal 4-1/2 gallon container, 12 inches in diameter and 12 inches high. Into this container, a cylindrical specimen rack and air distribution manifold shall be placed as shown on figure 9. The shelves shall be of 1/4-inch bakelite and have bakelite separators to serve as legs and as spacers. Each shelf shall be perforated with numerous 1/2 inch diameter holes. Air at the rate of 0.3 cubic feet per minute shall be bubbled through a water column approximately 6 inches high and introduced into the solution through a manifold and interlaced ring assembly. The ring assembly shall be made of 1/4-inch diameter plastic tubing and have a total of fifty-one 3/64-inch diameter holes. (It is to be noted that no metal is used in any part of this apparatus.)

4.6.11.2 Procedure. Three 2-inch by 6-inch and three 6-inch by 6-inch specimens shall be prepared as specified in 4.6.1. All specimens shall be coated on all sides with one spray coat of epoxy primer, conforming to MIL-P-24441/1, formula 150, at approximately 3 mils dry film thickness, except for a 1/2-inch wide strip of exposed metal at the end of one face of each of the 2-inch by 6-inch specimens. Adjacent to the exposed metal, a 1-inch strip of primer shall not be covered with a topcoat. The 2-inch by 4-inch remainder of the face shall have the complete system of primer and topcoat. The three coated 6-inch by 6-inch specimens shall be subjected to impact as specified in 4.6.7.2, prior to immersion in synthetic

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seawater. The specimens shall be placed on the shelves with the coating system face up and the glass jar filled to within approximately 3 inches from the top with synthetic seawater prepared from sea salt in accordance with ASTM D-1141-52. The specimen shall be immersed in the seawater for 15 days, during which a continuous stream of air shall be passed through the water to increase the severity of the corrosion environment.

4.6.12 Resistance to moisture and temperature changes. Tests shall be made on three 2-inch by 6-inch specimens prepared as specified in 4.6.1 and 4.6.11.2. Each specimen shall be immersed for 48 hours in synthetic seawater prepared from sea salt in accordance with ASTM D-1141-52. Immersion shall be under a pressure of 8 pounds per square inch. Immediately after immersion, the specimen shall be subjected to two complete cycles of alternate exposure to a temperature of plus or minus 5°F, for 24 hours, followed by a temperature of 120 plus or minus 5°F for 24 hours. The specimen shall then be examined for evidence of cracking or other failure. A portion of the coating shall be carefully removed from the steel plate to observe any signs of rusting or corrosion under the coating.

4.6.13 Resistance to immersion. Twenty-one 2-inch by 6-inch specimens shall be prepared as specified in 4.6.1 and 4.6.11.2. Three specimens shall be positioned vertically with respect to their 6-inch length and exposed for 24 hours to one of the seven media listed below. The immersed portion of the specimen shall include the strips of base and primed metal while the top 2 inches of each specimen shall remain unexposed for purposes of comparison.

Test media
(a) SAE 10W oil
(b) Grease conforming to MIL-G-18709
(c) Aviation gasoline 115/145 conforming to MIL-F-5572
(d) Jet fuel JP-5 conforming to MIL-T-5624
(e) Hydraulic fluid conforming to MIL-H-5606
(f) Aircraft ethyl alcohol conforming to MIL-A-6091
(g) Detergent conforming to MIL-D-16791, 1/2 percent solution

4.6.13.1 Upon removal from the immersion medium, the deck covering systems shall be examined for softening, loss of adhesion at the primer-topcoat interfaces, or other forms of deterioration.

4.6.14 Fire resistance test. The material shall be tested in accordance with the test procedure set forth in MIL-STD-1623.

4.6.15 Flash point. The flash point shall be determined immediately after mixing in accordance with method 4293 of FED-STD-141.

4.6.16 Color. A specimen 6 by 6 inches, prepared as specified in 4.6.1, shall be compared with the standard color chip (see 3.20).

4.6.17 Serviceability and service use. A service test shall be conducted on the qualified product (see 4.3 and 4.3.2) on a carrier flight deck (landing area or non-landing area, as applicable), in the specific areas as defined by the Naval Ship Engineering Center. Material shall be applied at the coverage and thickness obtained in the laboratory qualification tests in accordance with the manufacturer's mixing and spraying instructions. For type I material, performance shall be determined at 6 months and 12 months to determine compliance with 3.22 and 3.23. For type II material, performance shall be determined after 1500 landings.

4.6.18 Inspection of preparation for delivery requirements. Packaging, packing, and marking shall be examined to determine conformance with the requirements of section 5, except sampling for examination of filled containers shall be in accordance with 4.4.1.1 and 4.4.2.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging.

5.1.1 Levels A and C. The topcoat shall be supplied as a kit, consisting of base material marked Component A, and curing agent marked Component B. The quantity of each component in the kit shall be in the proportions for curing approved under qualification. Component A shall be furnished in 1-gallon cans or pails or in 5-gallon pails as specified

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(see 6.2). Component B shall be furnished in cans or pails of appropriate size to contain the curing quantity for over 5 gallons of base resin. Alternatively, the topcoat kit, Components A and B, may be packaged in a 6-gallon pail provided with an auxiliary compartment.

5.1.1.1 Cans. Cans shall conform to type V, class 2 of PPP-C-96. Interior of cans may be unlined; plan B coating and side seam stripping are required. Wire handles treated to resist corrosion are required for 1-gallon cans.

5.1.1.2 One- and five-gallon pails. One- and five-gallon pails shall conform to type II, class 3 of PPP-P-704. Interior coating is not required. Wire handles or bails shall be treated to resist corrosion.

5.1.1.3 Double-compartment pail.

5.1.1.3.1 Six-gallon pail. The 6-gallon pail shall conform to type II class 5 or 12 at the supplier's option of PPP-P-704. Interior coating is not required. Wire handles or bails shall be treated to resist corrosion.

5.1.1.3.2 Auxiliary compartment. The auxiliary compartment may be a tray or of saddle configuration; fabricated of metal or plastic and designed to rest over the pail edge. Compartments shall be provided with a gasket(s) to provide a snug fit and leak-proof closure when pail lid is secured. The compartment shall be of sufficient size to accommodate the packaged quantity of curing agent, component B.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A and B.

5.2.1.1 Cans. Each kit packaged as specified shall be packed in a fiberboard box conforming to PPP-B-636, class weather resistant for level A and class domestic for level B. Fiberboard pads shall be placed on the bottom and tops of the cans. Box closure, waterproofing and reinforcing shall be in accordance with method V of the appendix to the box specification.

5.2.1.2 Pails. One-gallon pails shall be packed for level A and B as specified in 5.2.1.1. Pails require no further packing for levels A and B. When specified (see 6.2), pails shall be palletized in accordance with MIL-STD-147.

5.2.2 Level C. Deck covering, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Container packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required herein (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each component container shall have the following additional information:

(a) Specification number and type
(b) Manufacturer's name and QPL brand designation
(c) Manufacturer's lot number and date of manufacture
(d) Contract number
(e) Color designation and number
(f) Shelf-life marking

5.3.1.1 Mixing, spraying, and use instructions. Each component container shall be marked with the following (see 3.24):

CAUTION

This is one component of a two component system which
WILL NOT HARDEN unless both components are mixed together.

INSTRUCTIONS FOR USE

Refer to manufacturer's written mixing and spraying
instructions prior to use.

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5.3.1.1.1 Additional type I marking. For type I kit, containers shall be marked on lid and container with the following in addition to other requirements.

DO NOT USE IN CARRIER LANDING AREA

5.3.1.1.2 Air pollution compliance. For those products which have been approved for use in areas with regulations controlling the emission of solvents into the atmosphere (see 3.21 and 6.5), containers shall include the following statement. "Complies with air pollution regulations."

5.3.1.2 Caution label. Each component container shall be marked with the following:

CAUTION: Avoid skin contact during application.
Chemicals may cause irritation or skin sensitization.
In case of contact, wash skin thoroughly with soap
and water.

5.3.1.3 Interior containers shall be identified in accordance with FED-STD-313. For shipping containers and palletized unit loads, a copy of the Material Safety Data Sheet DD 1813 shall be attached to the shipping document for each destination. The contractor shall furnish a copy of the DD 1813 to the Naval Ship Engineering Center with application for product qualification.

5.4 Primer. The primer shall be prepared for shipment in accordance with the requirements of MIL-P-24441/1.

6. NOTES

6.1 Intended use. This material, when sprayed over a primed metal (or wood) deck, is intended to provide safe footing for both machinery and personnel under dry, oily, or wet conditions.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of deck covering (see 1.2).
(c) Color material (see 3.20).
(d) Compliance with air pollution regulations required (see table II).
(e) Number of copies of manufacturer's written directions for mixing and spraying (see 3.24).
(f) Quantity of kits and size of component containers (see 5.1).
(g) Type container for 1 gallon quantity: can or pail (see 5.1).
(h) Level of packaging and packing required (see 5.1 and 5.2).
(i) Palletization, when required (see 5.2.1.2).
(j) Special marking required (see 5.3).

6.2.2 Contract data requirements. When this specification is used in a procurement invoking the data requirements clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirements	Service	Applicable DID	Options
(a)	4.4.3	Data, Quality Assurance	SH	UDI-T-23314	---

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24483 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Formulations which comply with AIR POLLUTION regulations may be required for use in areas with regulations controlling the emission of solvents into the atmosphere.

6.6 Color chips may be obtained from Specification Sales (Code 3FRSBS), Building 197, Washington Navy Yard, General Services Administration, Washington, D.C. 20407.

6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 5610-N024)

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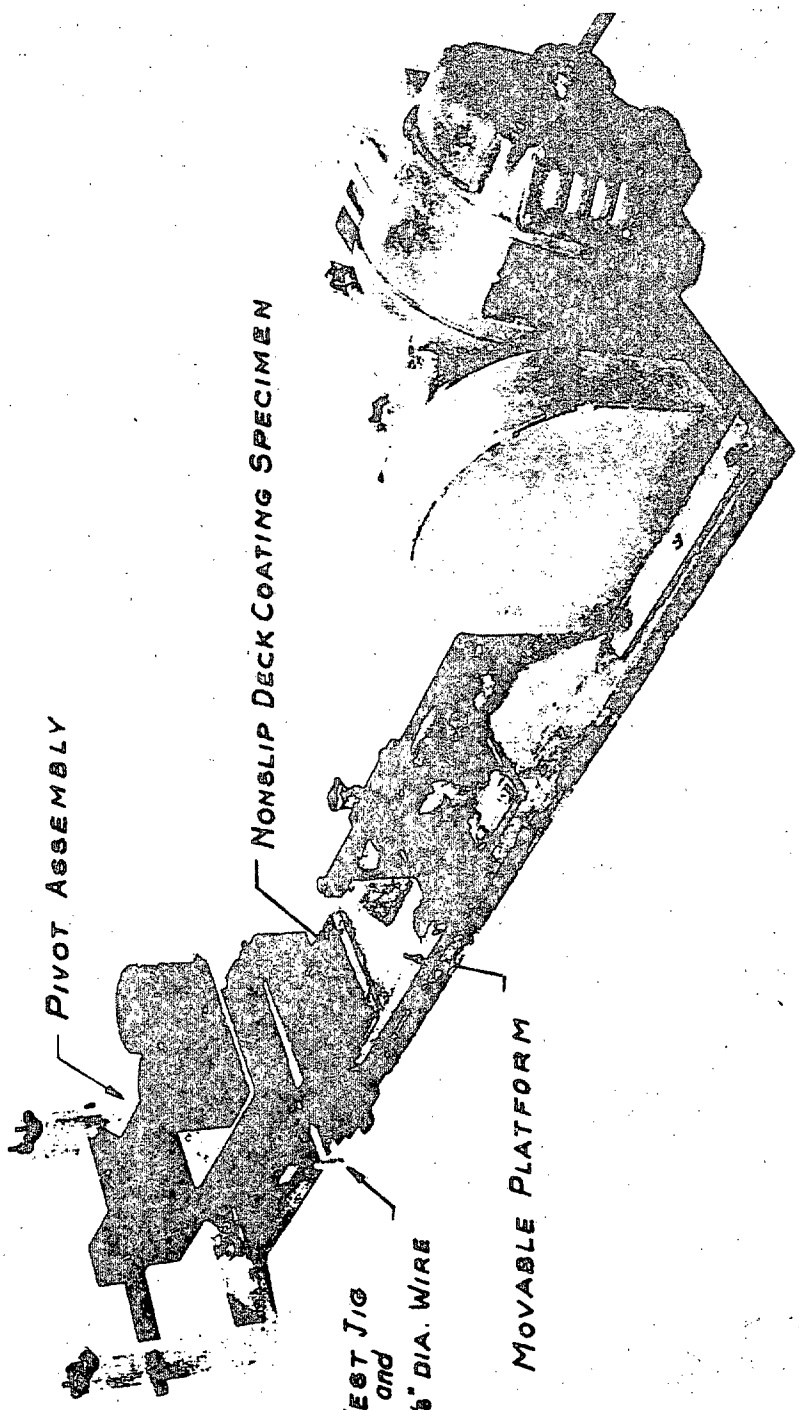
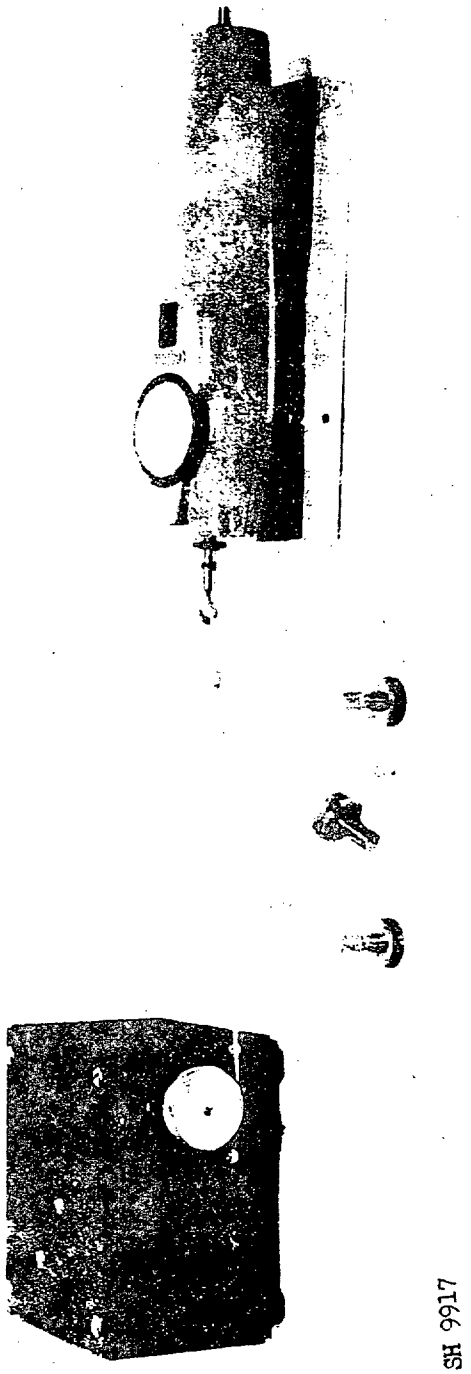


Figure 1 - Arresting gear cable wear test apparatus.

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SH 9917

Figure 2 - Coefficient of static friction apparatus.

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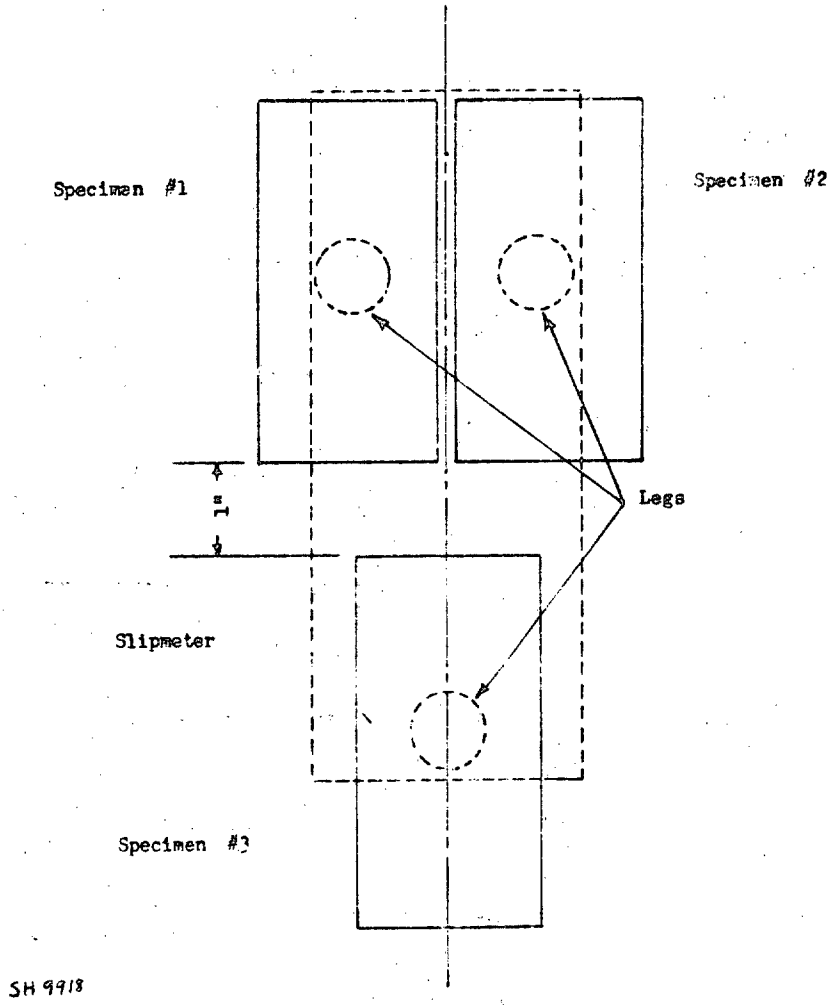


Figure 3 - Arrangement of 2 by 3 inch specimens for measurement of coefficient of friction.

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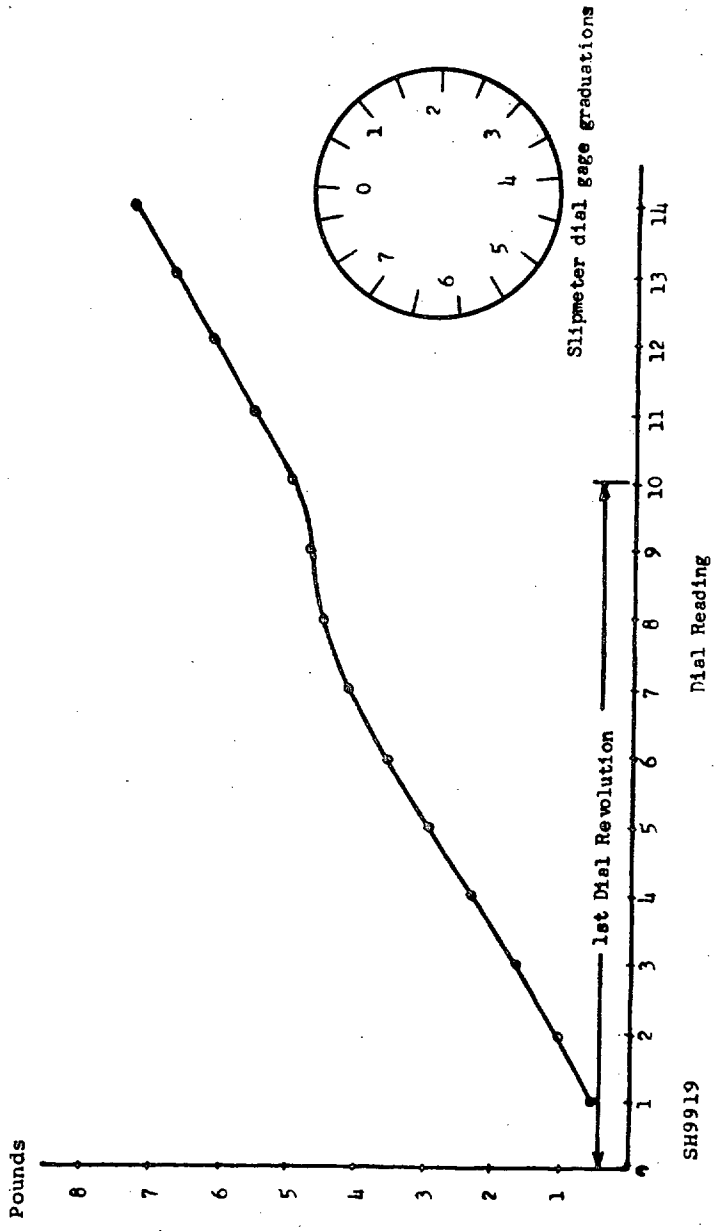


Figure 4 - Conversion curve of olsen slipmeter dial readings to load in pounds.

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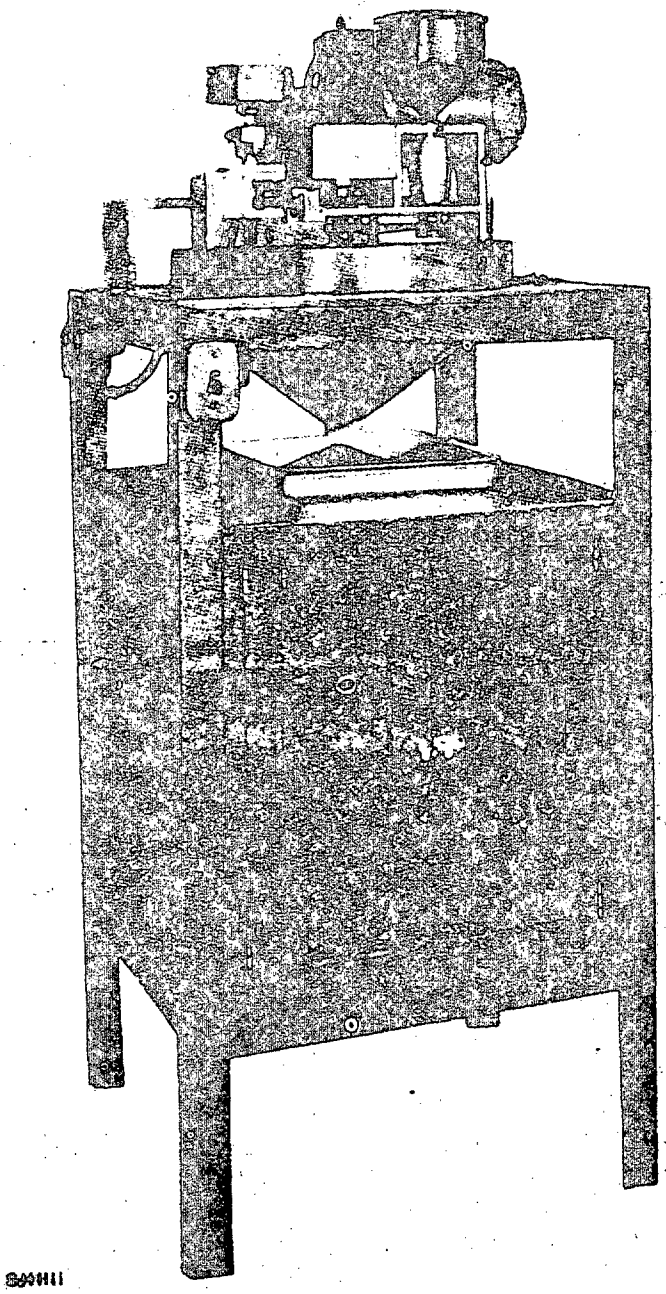


Figure 5 - Wear test machine.

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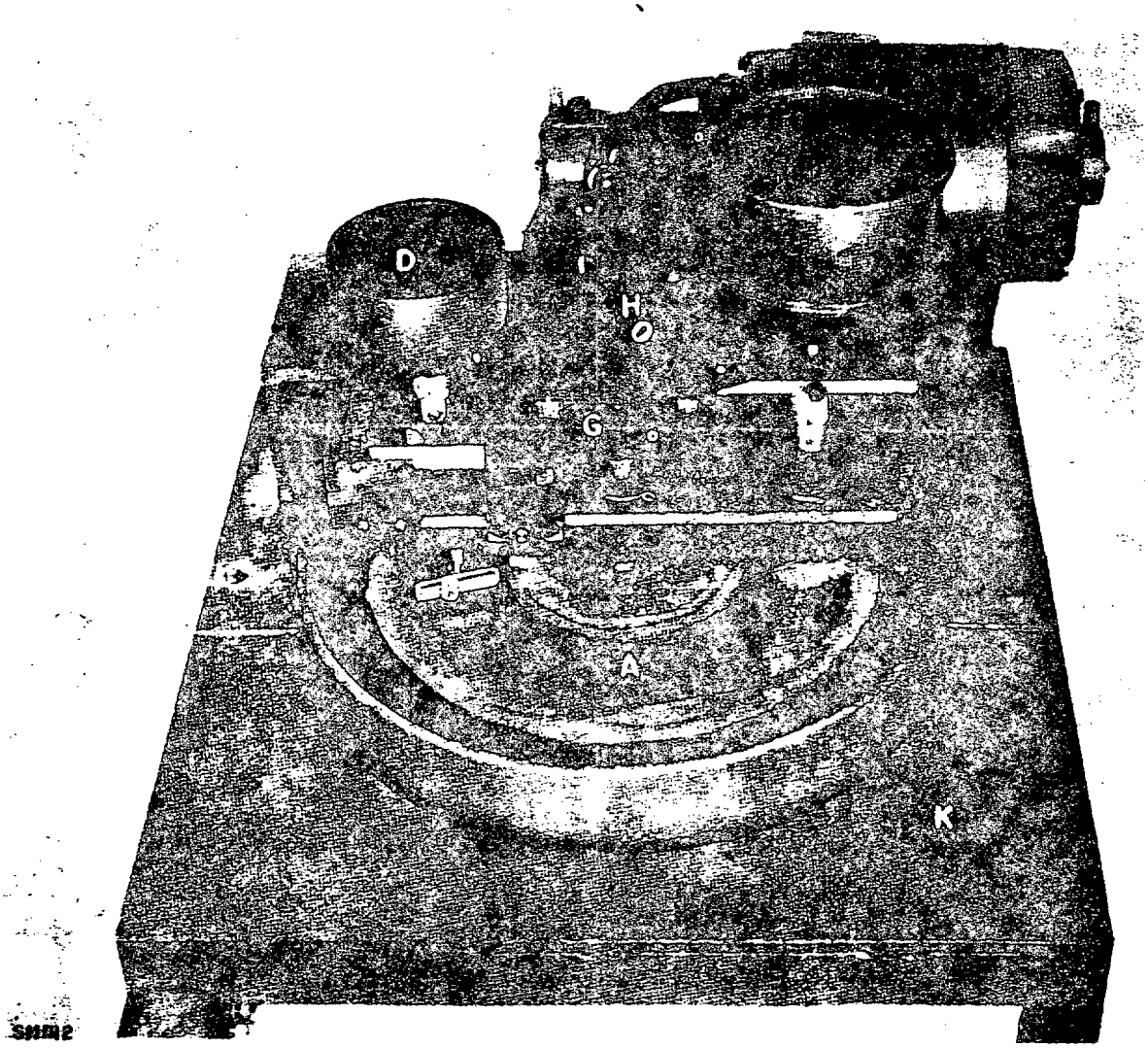


Figure 6 - Detail view of wear-test machine.

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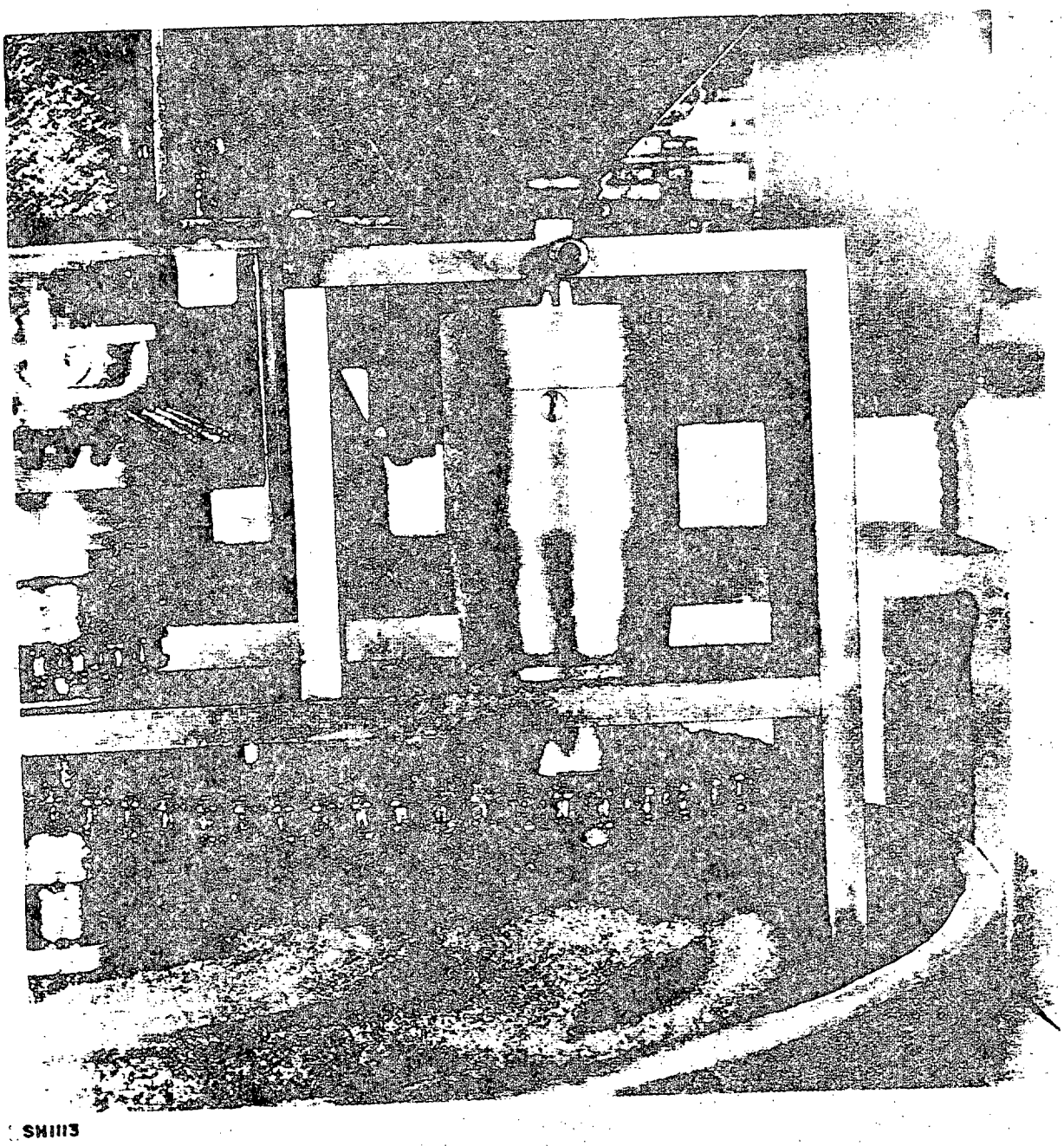
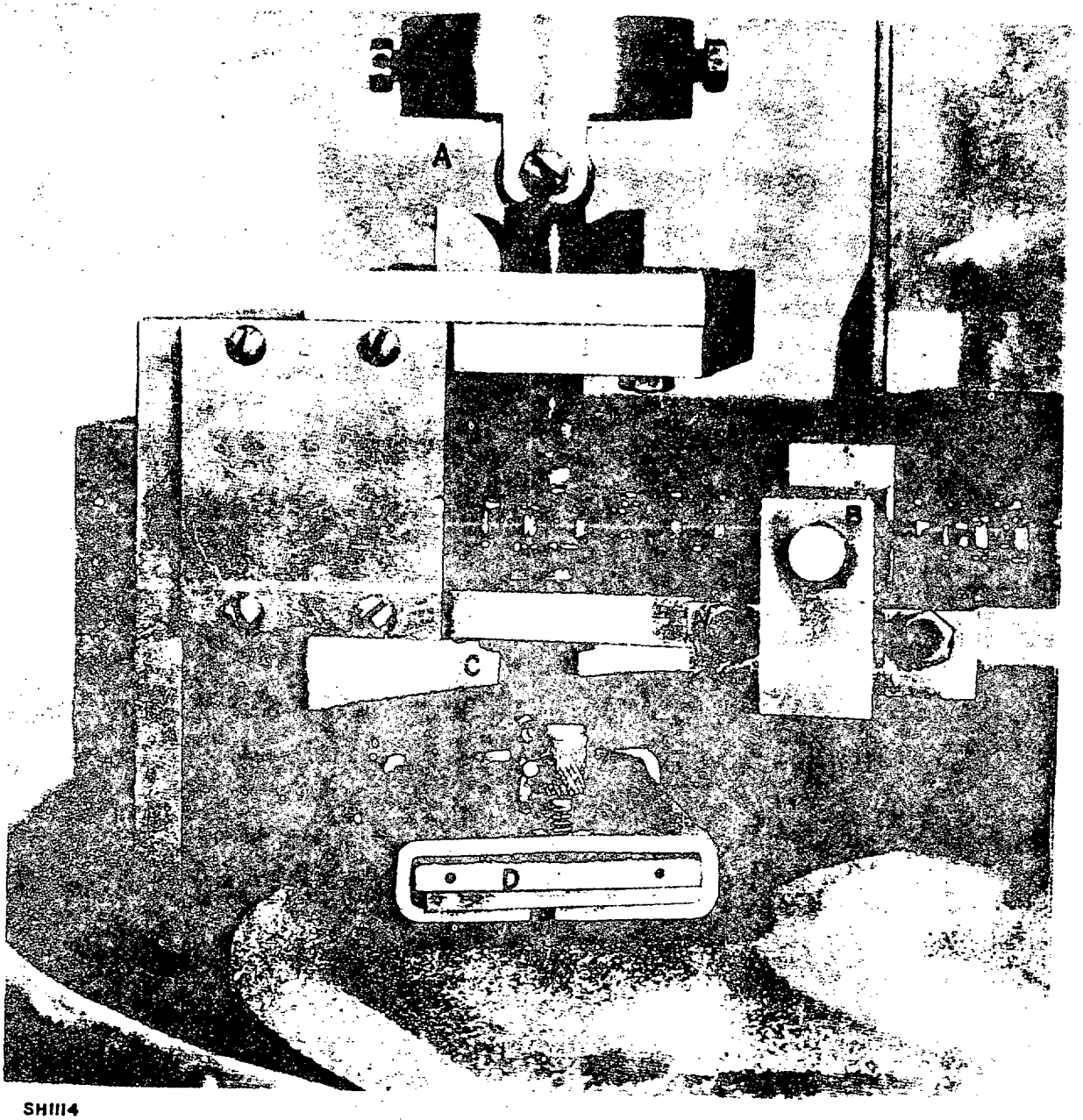


Figure 7 - Detail of distributor of abrasive grit.

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SH1114

Figure 8 - Detail of cam and specimen holder.

A - Cam
B - Chain idler
C - Universal joint
D - Specimen holder

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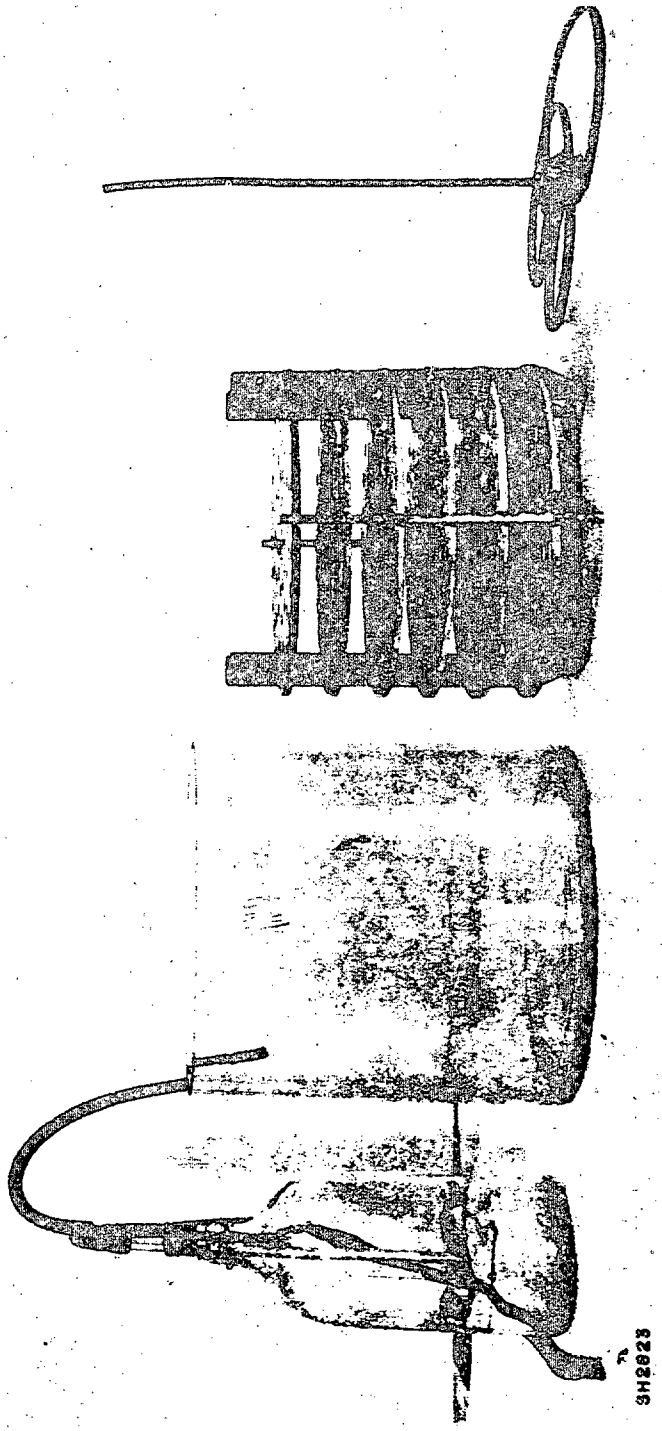


Figure 9 - Accelerated corrosion.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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